



## Related Work

CIF builds on and extends several research traditions. We position our contributions relative to the most closely related work in each area, highlighting both the foundations we draw upon and the novel elements that distinguish our approach.

### Prompt Injection Defenses

The growing body of work on prompt injection defenses has largely focused on single-agent scenarios. Greshake et al. [?] demonstrated indirect prompt injection through tool outputs in LLM-integrated applications, revealing how malicious content in retrieved documents can hijack agent behavior. Perez and Ribeiro [?] characterized injection vulnerabilities through large-scale competitive testing, establishing benchmark datasets for injection detection. Liu et al. [?] provided a taxonomy of injection attacks against LLM applications, categorizing techniques by attack vector and target. More recently, the *Prompt Infection* paradigm [?] has demonstrated that injections can self-replicate across LLM agents: a compromised agent's output embeds injection payloads that propagate to downstream agents, creating epidemic-like attack